Machine Learning (69152)

Bayesian Inference

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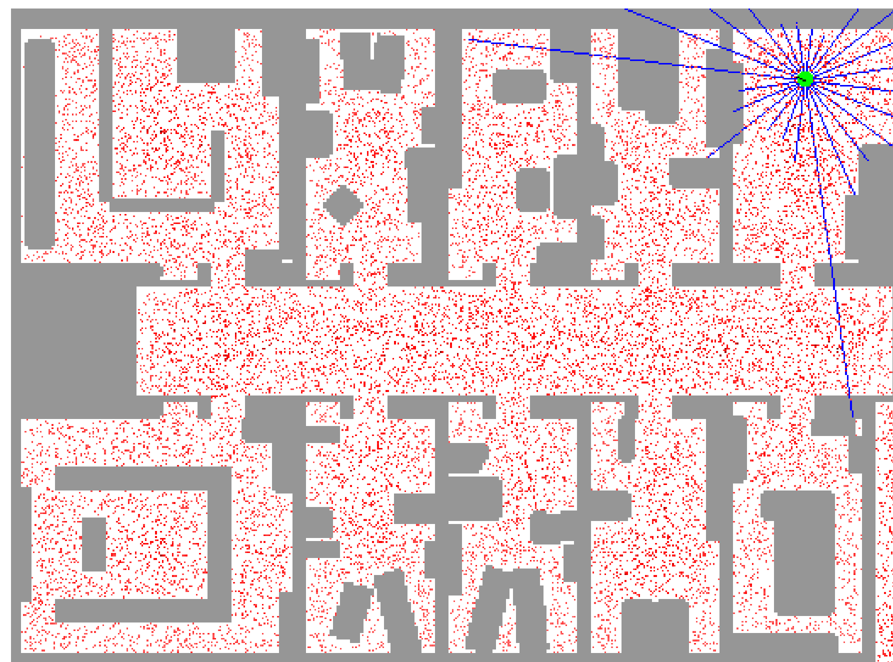
Monte Carlo methods

Examples



Bayesian logistic regression

- The logistic regression model represents the probability of a binary output $y_i \in \{0, 1\}$ given the input $\mathbf{x}_i$:

$$p(\mathbf{y}|\mathbf{X}, \theta) = \prod_{i=1}^{n} Ber(y_i|sigm(\theta^T \mathbf{x}_i)) = \prod_{i=1}^{n} \left[\frac{1}{1 + e^{-\theta^T \mathbf{x}_i}}\right]^{y_i} \left[1 - \frac{1}{1 + e^{-\theta^T \mathbf{x}_i}}\right]^{1-y_i}$$

- We also assume a Gaussian prior:

$$p(\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} exp\left(-\frac{1}{2\sigma^2}(\theta - \mu)^T(\theta - \mu)\right)$$

- What is the posterior? What is the predictive posterior?

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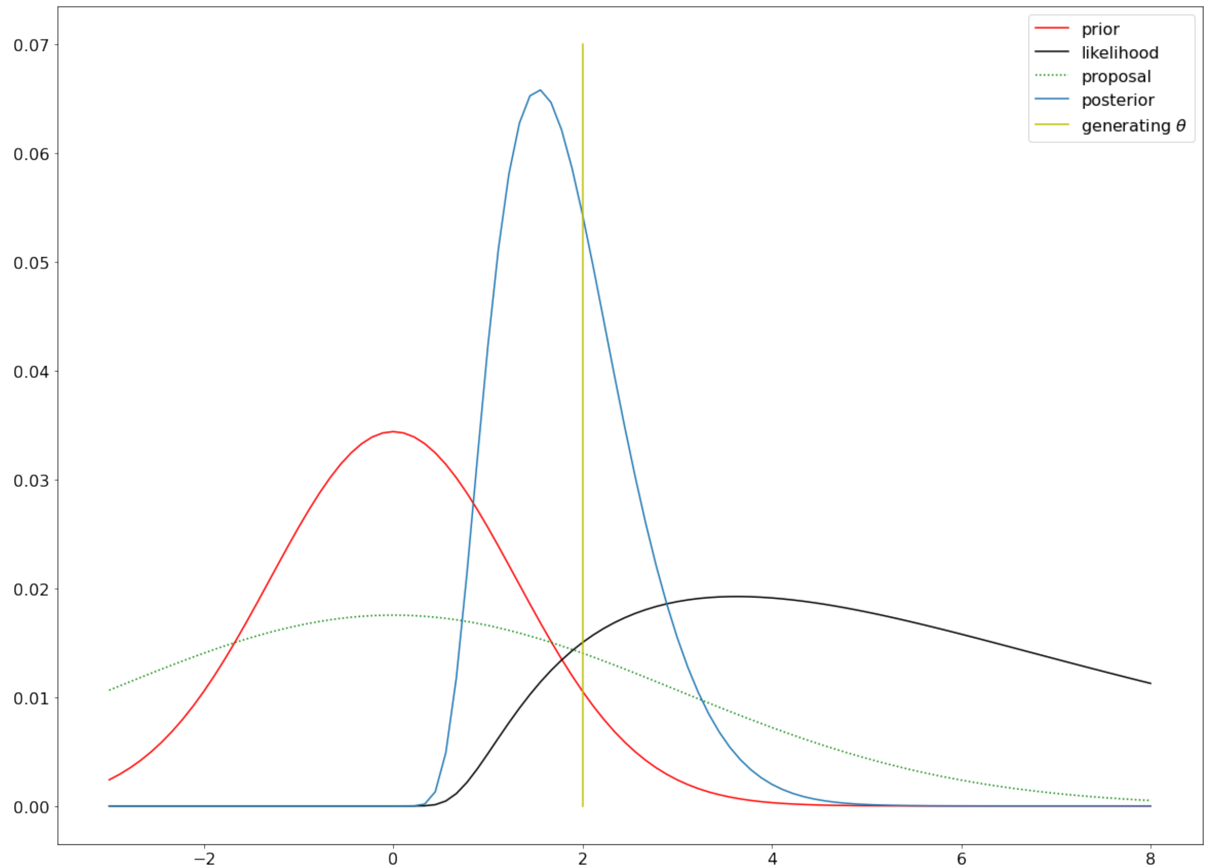
Bayesian logistic regression

$$p(\theta|\mathbf{X},\mathbf{y}) = \frac{p(\mathbf{y}|\mathbf{X},\theta)p(\theta)}{\int p(\mathbf{y}|\mathbf{X},\theta)p(\theta)d\theta} = \frac{1}{Z}p(\mathbf{y}|\mathbf{X},\theta)p(\theta)$$

$$p(y_*|\mathbf{x}_*,\mathbf{X},\mathbf{y}) = \int p(y_*|\mathbf{x}_*,\theta)p(\theta|\mathbf{X},\mathbf{y})d\theta$$



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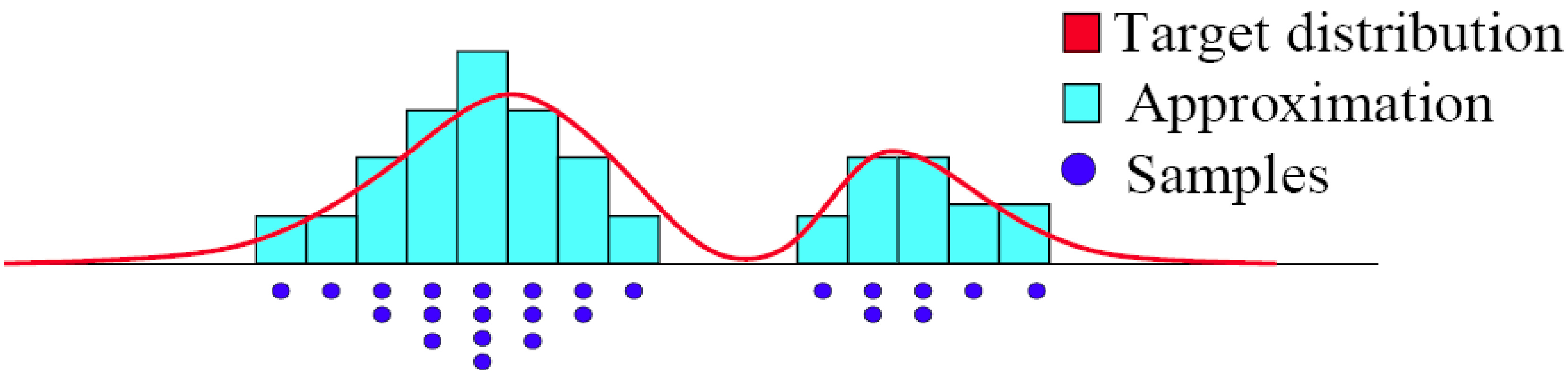
Monte Carlo Principle

- Approximate a distribution

$$p(x) = \frac{1}{N}\sum_{i=1}^{N} \delta_{x^{(i)}}(x)$$



$$I_N(f) = \frac{1}{N}\sum_{i=1}^{N} f(x^{(i)}) \xrightarrow[N \to \infty]{a.s.} I(f) = \int_{\mathcal{X}} f(x)p(x)dx$$

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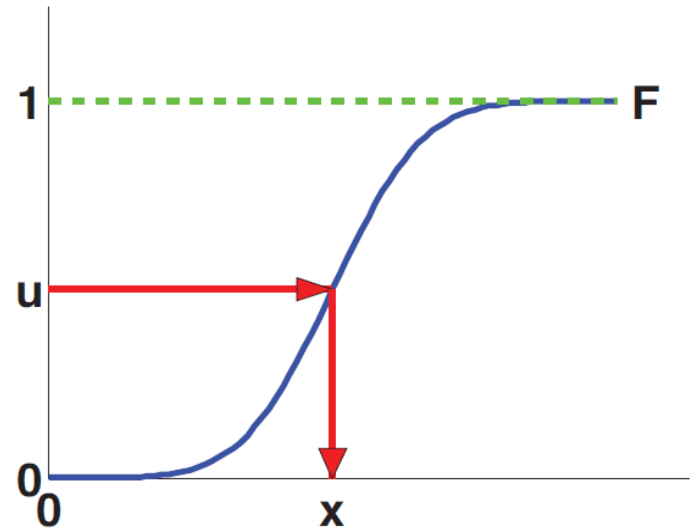
Proposal distribution

- Sampling from a distribution
  - Inverse cumulative distribution "trick"
  - This can be done only for certain distributions
    - Gaussian
    - Uniform
    - …



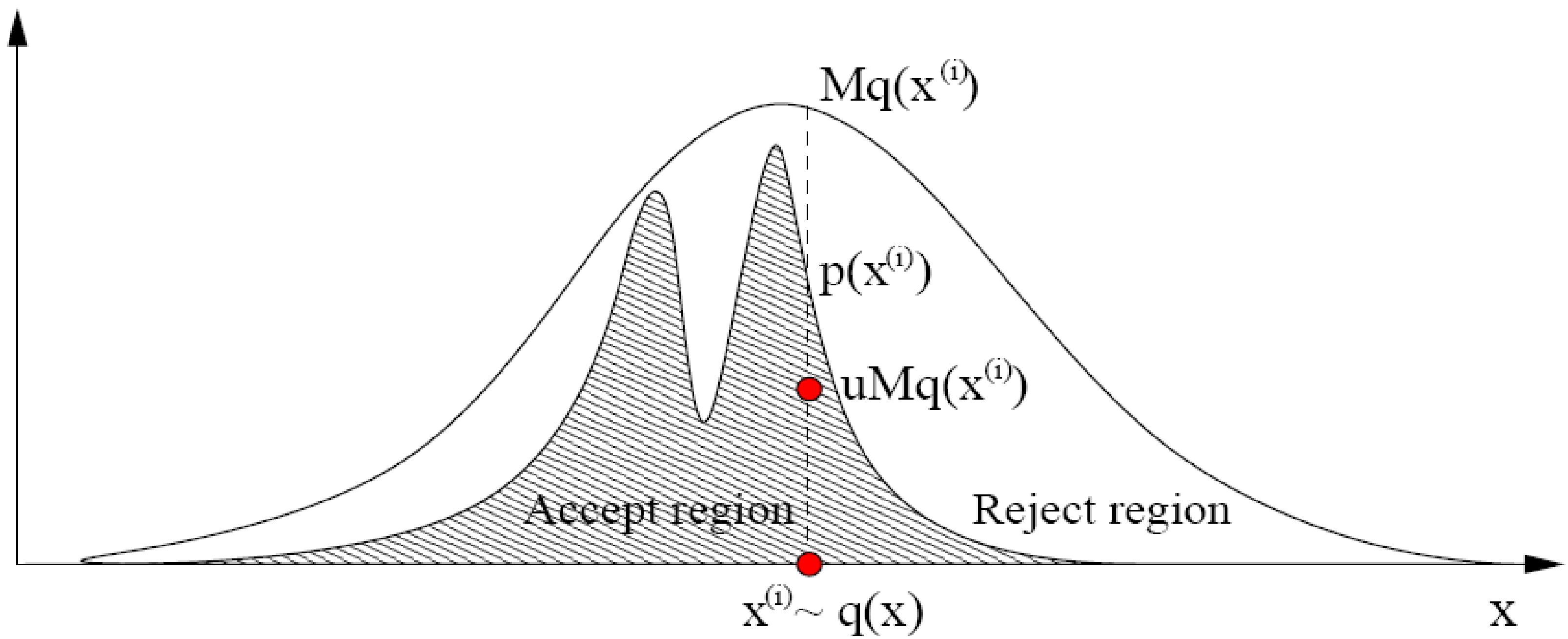
- Solution: use a different distribution for sampling (proposal) and for evaluating (target)

Rejection Sampling

- The *proposal* q(x) must be greater than the *target* p(x)

$$Mq(x) \geq p(x) \qquad \forall x \in \mathbb{R}^d$$

- R



Video: Kevin P. Murphy

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Importance Sampling

- Can we use a more general proposal q(x)?

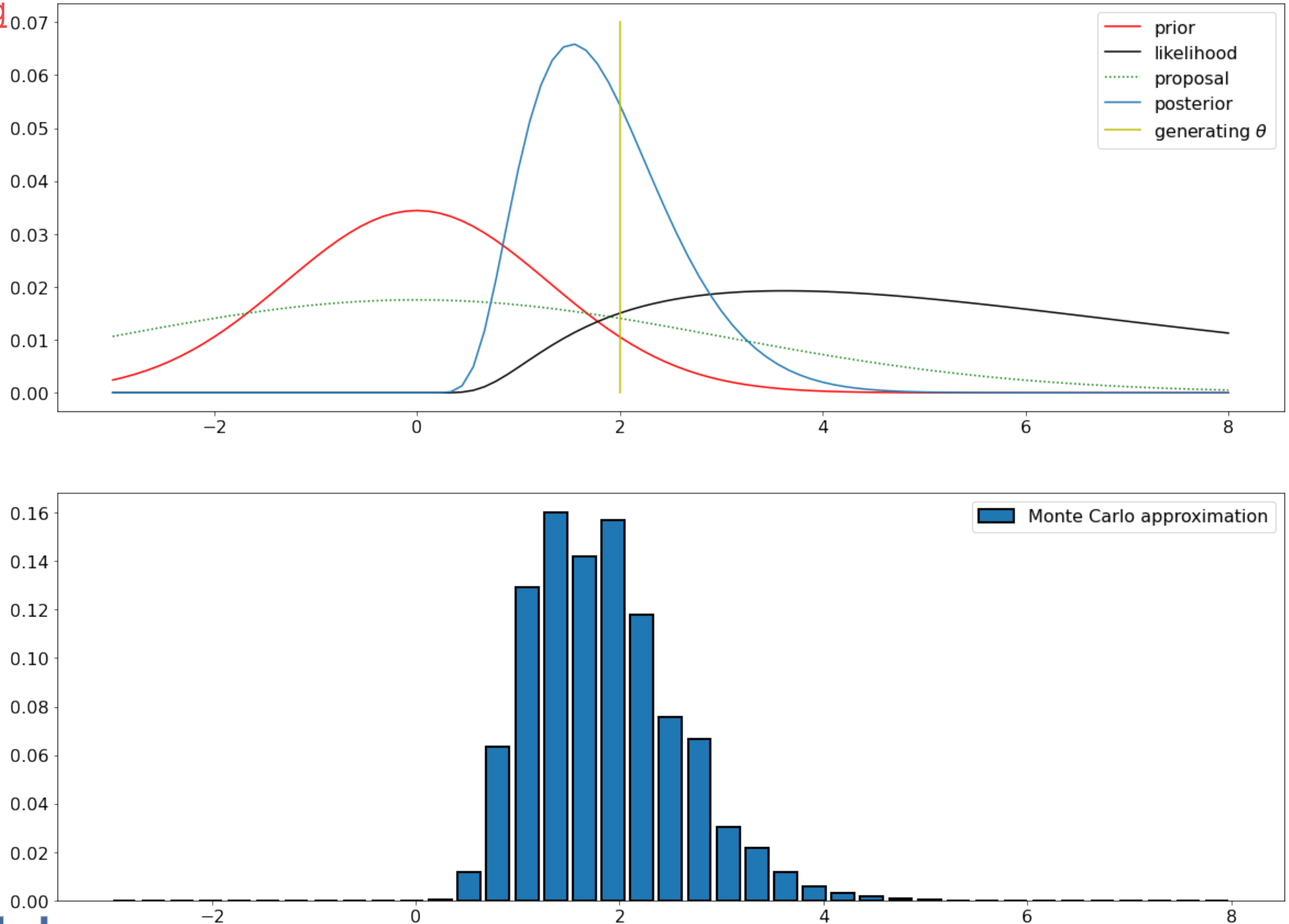
$$
\begin{aligned}
I(f) &= \int f(x)p(x)dx \\
&= \int f(x)\frac{p(x)}{q(x)}q(x)dx \\
&= \int f(x)w(x)q(x)dx
\end{aligned}
$$

- Thus, the Monte Carlo estimate becomes

$$\hat{I}_N(f) = \sum_{i=1}^{N} f(x^{(i)})w(x^{(i)})$$

Importance sampling

https://colab.research.google.com/drive/1AJmsrr6fWRro1GjYtfKRVpMBhXmKTzm6?usp=sharing



Un-normalized importance sampling

$$p(\theta|\mathbf{X},\mathbf{y}) = \frac{p(\mathbf{y}|\mathbf{X},\theta)p(\theta)}{\int p(\mathbf{y}|\mathbf{X},\theta)p(\theta)d\theta} = \frac{1}{Z}p(\mathbf{y}|\mathbf{X},\theta)p(\theta)$$

$$p(y_*|\mathbf{x}_*,\mathbf{X},\mathbf{y}) = \int p(y_*|\mathbf{x}_*,\theta)p(\theta|\mathbf{X},\mathbf{y})d\theta = \frac{1}{Z}\int p(y_*|\mathbf{x}_*,\theta)p(\mathbf{y}|\mathbf{X},\theta)p(\theta)d\theta$$

$$= \frac{1}{Z}\int p(y_*|\mathbf{x}_*,\theta)\frac{p(\mathbf{y}|\mathbf{X},\theta)p(\theta)}{q(\theta)}q(\theta)d\theta = \frac{1}{Z}\int p(y_*|\mathbf{x}_*,\theta)w(\theta)q(\theta)d\theta \approx \frac{1}{Z}\frac{1}{N}\sum_{i=1}^{N} w(\theta^i)p(y_*|\mathbf{x}_*,\theta^i)$$

$$Z = \int p(\mathbf{y}|\mathbf{X},\theta)p(\theta)d\theta = \int \frac{p(\mathbf{y}|\mathbf{X},\theta)p(\theta)}{q(\theta)}q(\theta)d\theta = \int w(\theta)q(\theta)d\theta \approx \frac{1}{N}\sum_{i=1}^{N} w(\theta^i)$$

$$p(y_*|\mathbf{x}_*,\mathbf{X},\mathbf{y}) \approx \frac{\cancel{\frac{1}{N}}\sum_{i=1}^{N} w(\theta^i)p(y_*|\mathbf{x}_*,\theta^i)}{\cancel{\frac{1}{N}}\sum_{i=1}^{N} w(\theta^i)} = \sum_{i=1}^{N}\tilde{w}^i p(y_*|\mathbf{x}_*,\theta^i)$$

$$\boxed{\tilde{w}^i = \frac{w(\theta^i)}{\sum_{i=1}^{N} w(\theta^i)}}$$

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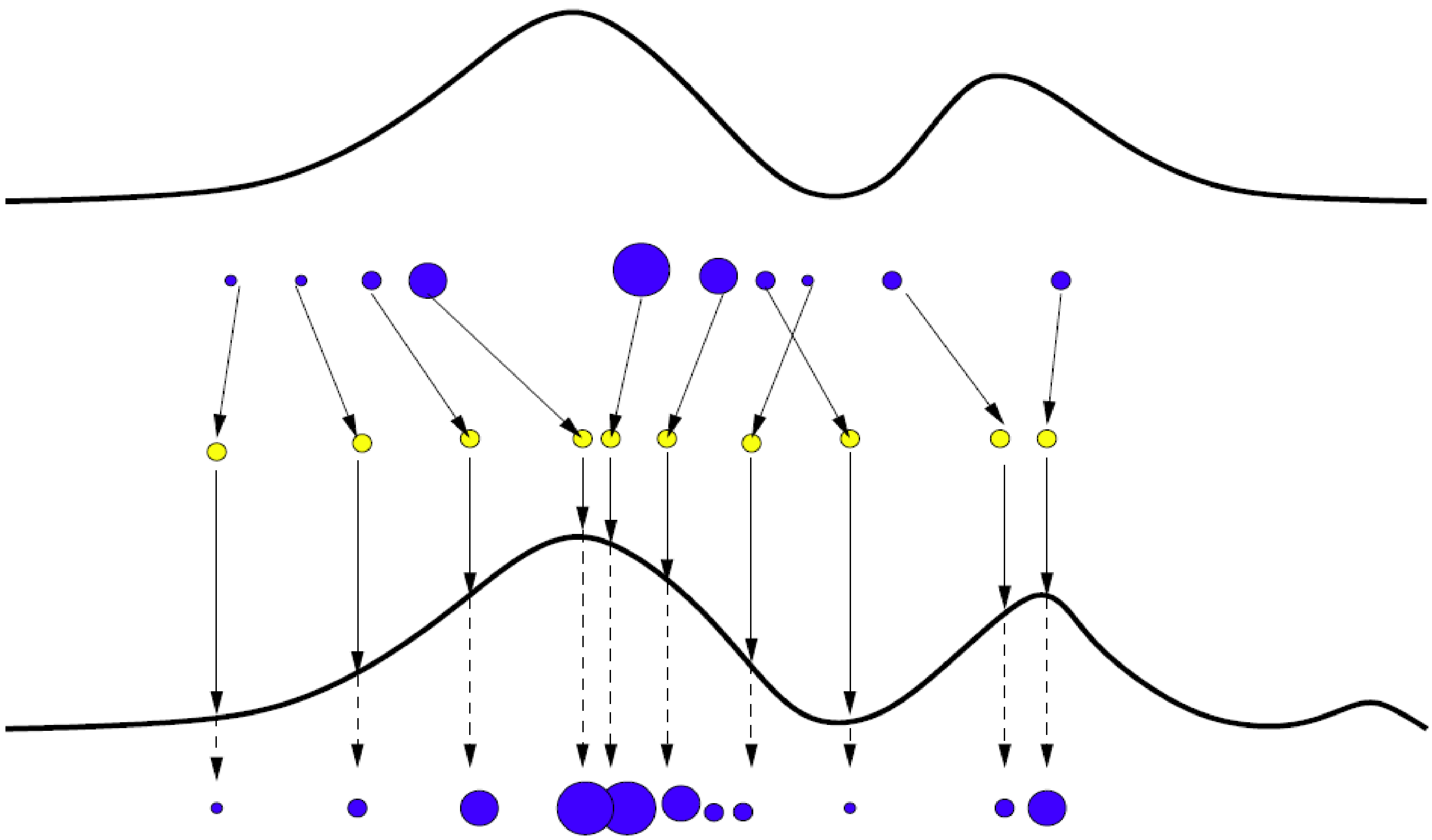
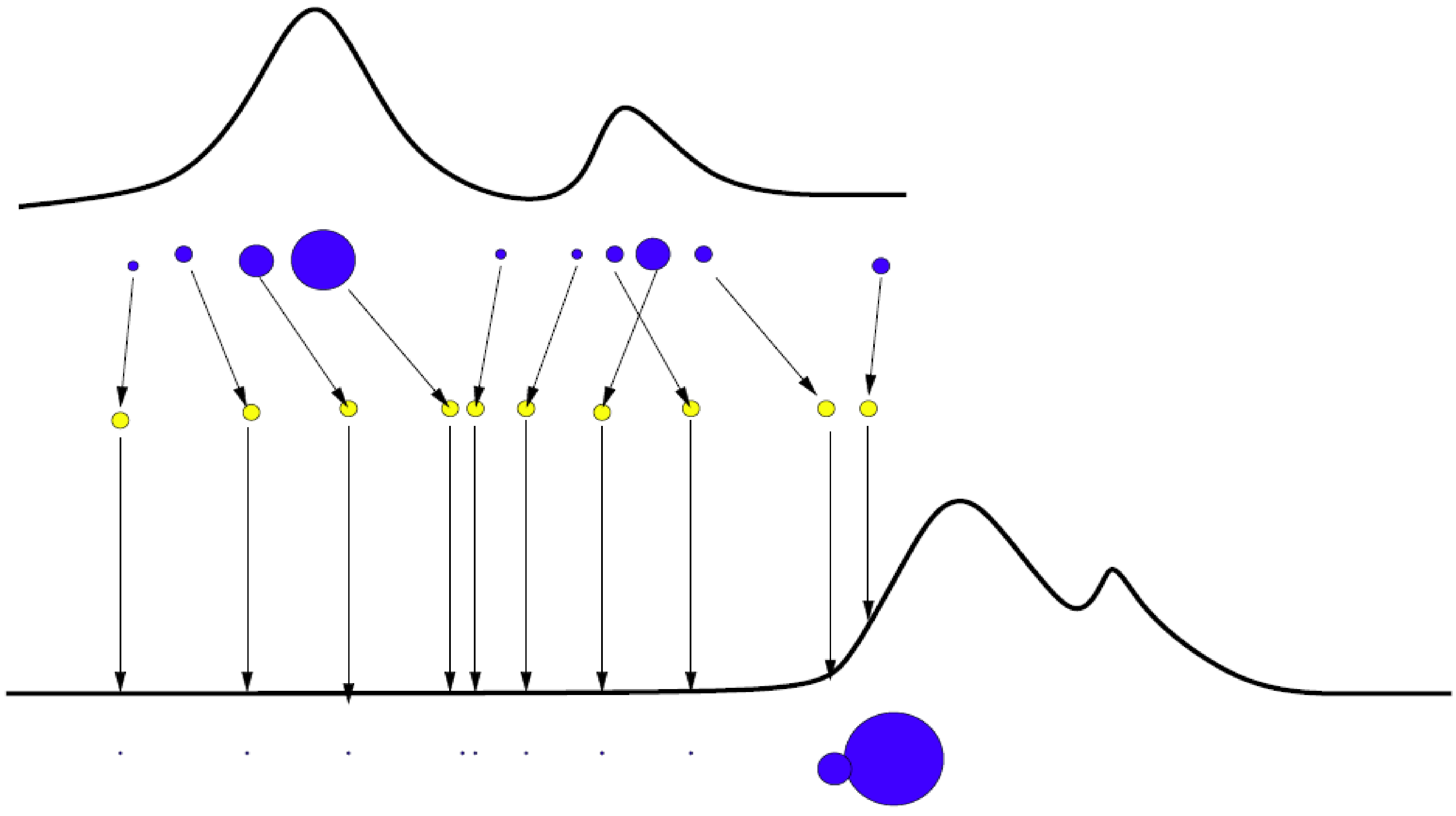
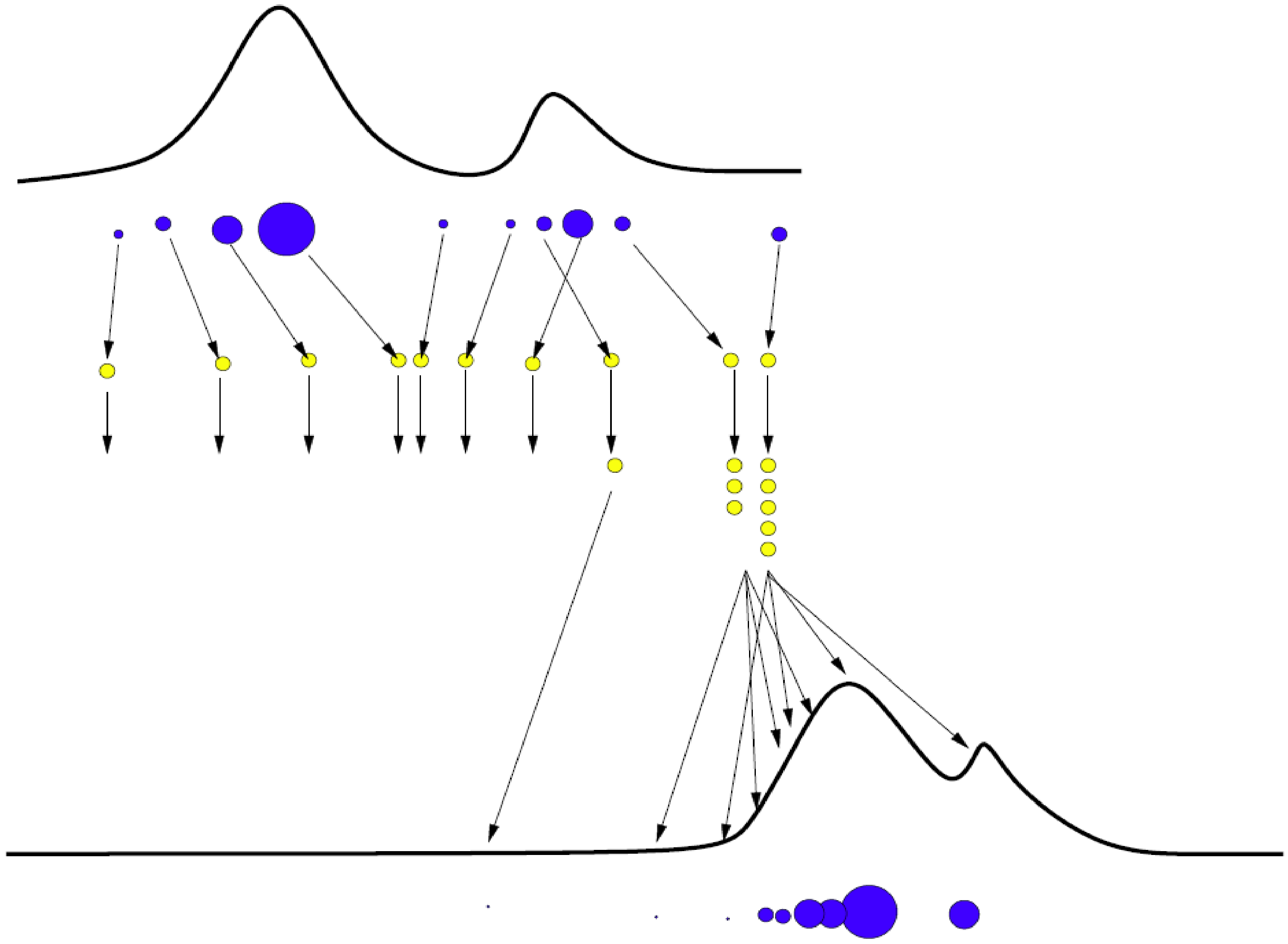
Sequential Importance Sampling (SIS)



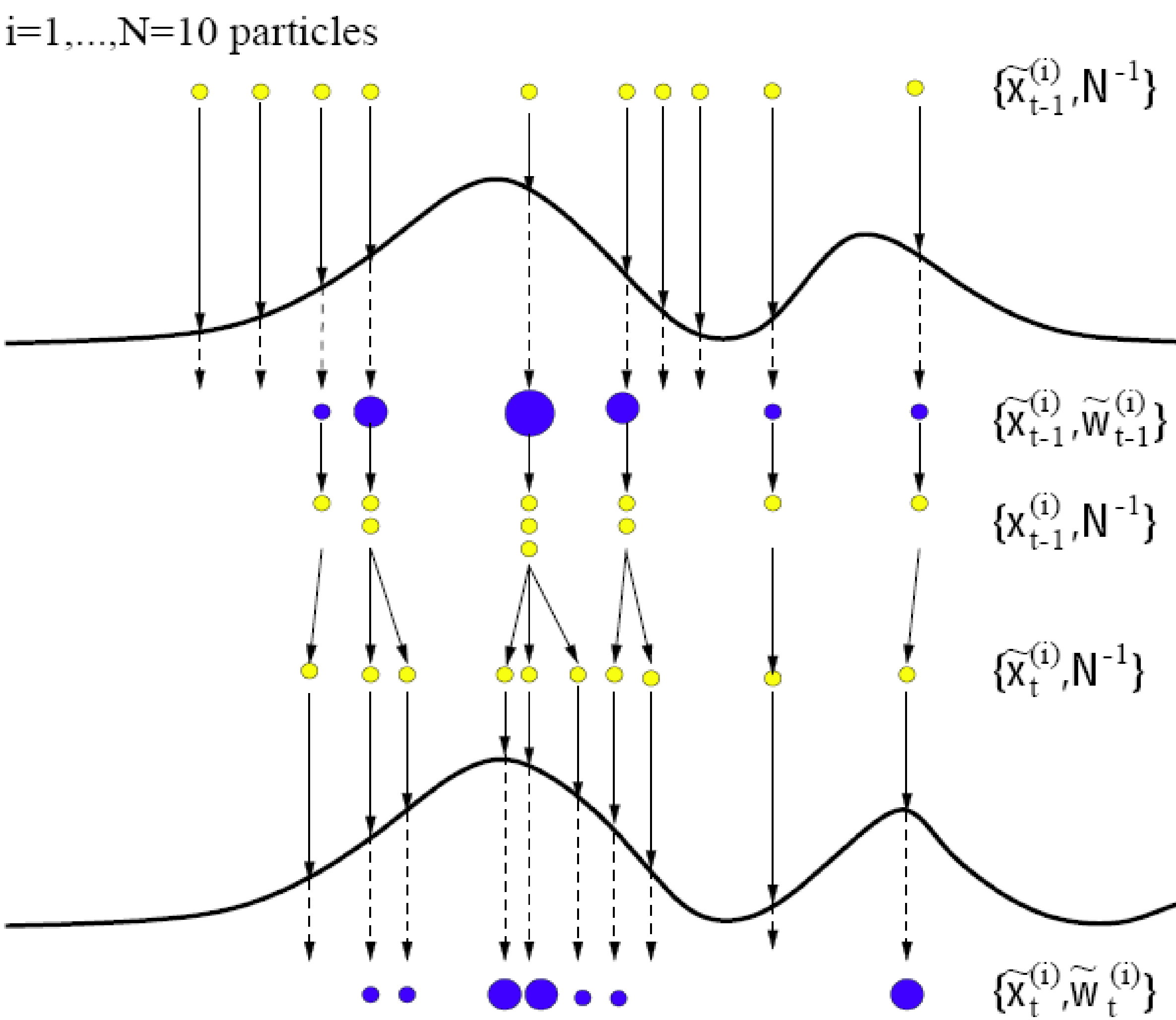
Sequential Importance Sampling (SIS)



SIS with Resampling (SIR)



Particle Filter



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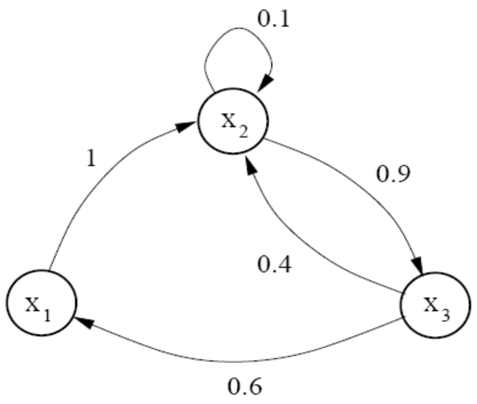
Markov Chains

- Let's consider only 3 states $x_t \in \mathcal{X} = \{x_1, x_2, x_3\}$
- T is the transition matrix

$$T = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0.1 & 0.9 \\ 0.6 & 0.4 & 0 \end{bmatrix}$$



- As soon as the chain is irreductible and aperiodic we have that for any initial probability vector $\nu(x)$:

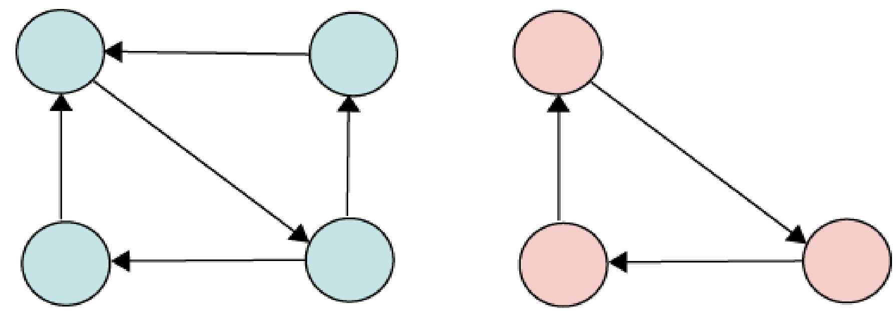
$$\nu T^t \to \pi \quad as \quad t \to \infty$$

where π is the stationary distribution (unique).

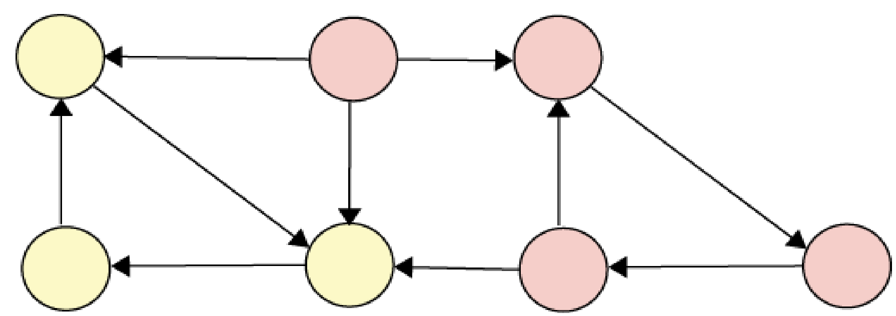
$$p(x) = (0.2, 0.4, 0.4)$$

Stability conditions

- Irreductibility: Connected.



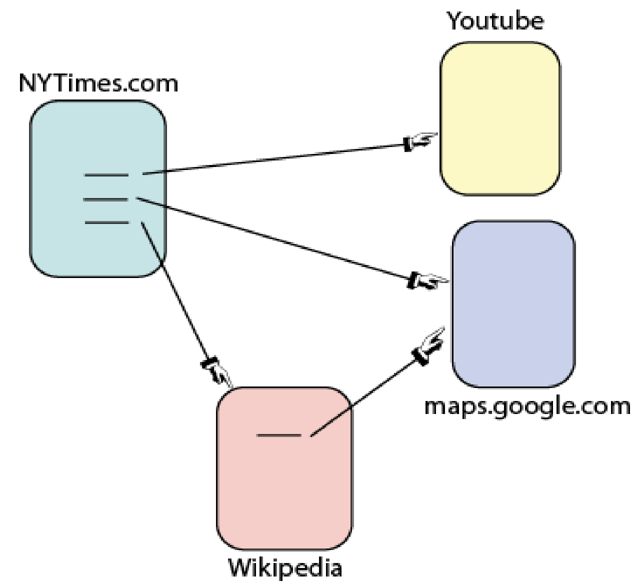
- Aperiodicity: No traps or end states.



Ordering massive data

Google

- Example: Pagerank
  - Represents the quality of a webpage.
  - It is the invariant distribution of the Markov chain of links.
  - We teach Google.



- How do we guarantee stability?
  - Add some random jumps

$$T = L + E$$

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Continuous MCMC

- In the limit:

$$\pi T = \pi$$

- The stationary distribution is the eigenvector of T with eigenvalue 1.
- We can write:

$$\sum_{i=1}^{3} \pi_i T_{ij} = \pi_j$$

- For continuous spaces, the transition matrix becomes a kernel $p(y|x)$.

$$\int \pi(x)p(y|x)dx = \pi(y)$$

Detailed Balance

- How do we know that the stable distribution is the distribution that we want?

- We design the kernel to enforce *detailed balance* (reversible).

$$\pi(x_t)p(x_{t+1}|x_t) = \pi(x_{t+1})p(x_t|x_{t+1})$$

- Which integrating over xt results in our desired behaviour

$$\int \pi(x_t)p(x_{t+1}|x_t)dx_t = \pi(x_{t+1})$$

- See algo Murphy2013 Sec 24.3.6

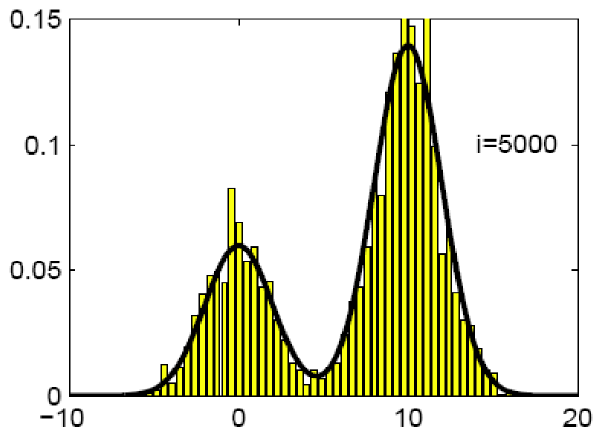
Design a MCMC algorithm

- We need to find an algorithm that:
  - Has detailed balance
    - It is irreducible and aperiodic

$$\pi(x_t)p(x_{t+1}|x_t) = \pi(x_{t+1})p(x_t|x_{t+1})$$

  - The stationary distribution is our target distribution

$$p(x) = \pi(x) \qquad \forall x$$



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Metropolis-Hasting

- Initialize $x^{(0)}$

- For i = 0 to N-1

  - Sample $x' \sim q(x'|x^t)$

  - Compute acceptance $r = \min\left(1, \frac{p(x')q(x^t|x\prime)}{p(x^t)q(x'|x^t)}\right)$

  - Sample $u \sim U(0,1)$

  - Set new sample $x^{t+1} = \begin{cases} x' & if\ u < r \\ x^t & if\ u \geq r \end{cases}$

MH has detailed balance

- If we analize the previous algorithm (call $x = x^t$):

$$p(x'|x) = \begin{cases} \boxed{q(x'|x)r(x'|x)} & \mathit{if}\ x' \neq x \\ \boxed{q(x|x)} + \boxed{\sum_{x' \neq x} q(x'|x)(1 - r(x'|x))} & \mathit{otherwise} \end{cases}$$

Jump accepted

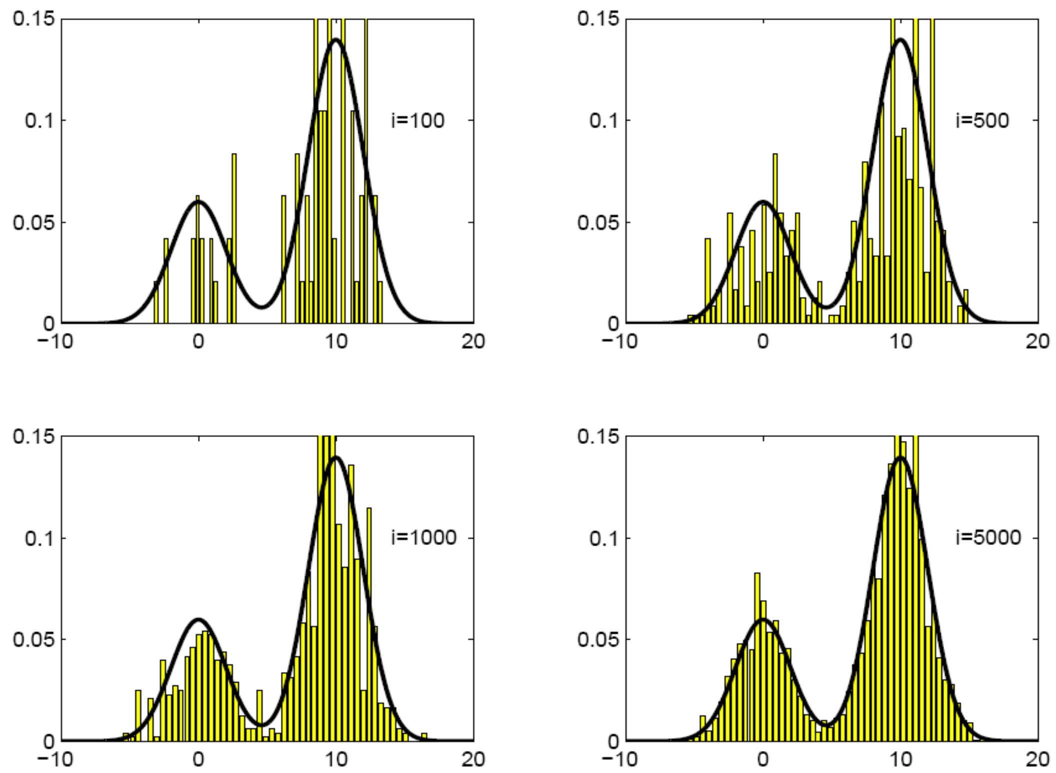
Jump to the same place

Jump rejected

- This has detailled balance and the stationary distribution is our target distribution (proof: Murphy2012 Sec 24.3.6)

MH Demos

- https://chi-feng.github.io/mcmc-demo/
- https://drive.google.com/file/d/1w9fPZMuy4T5OAaETS5q6vooj7SxL4jp4/view?usp=sharing



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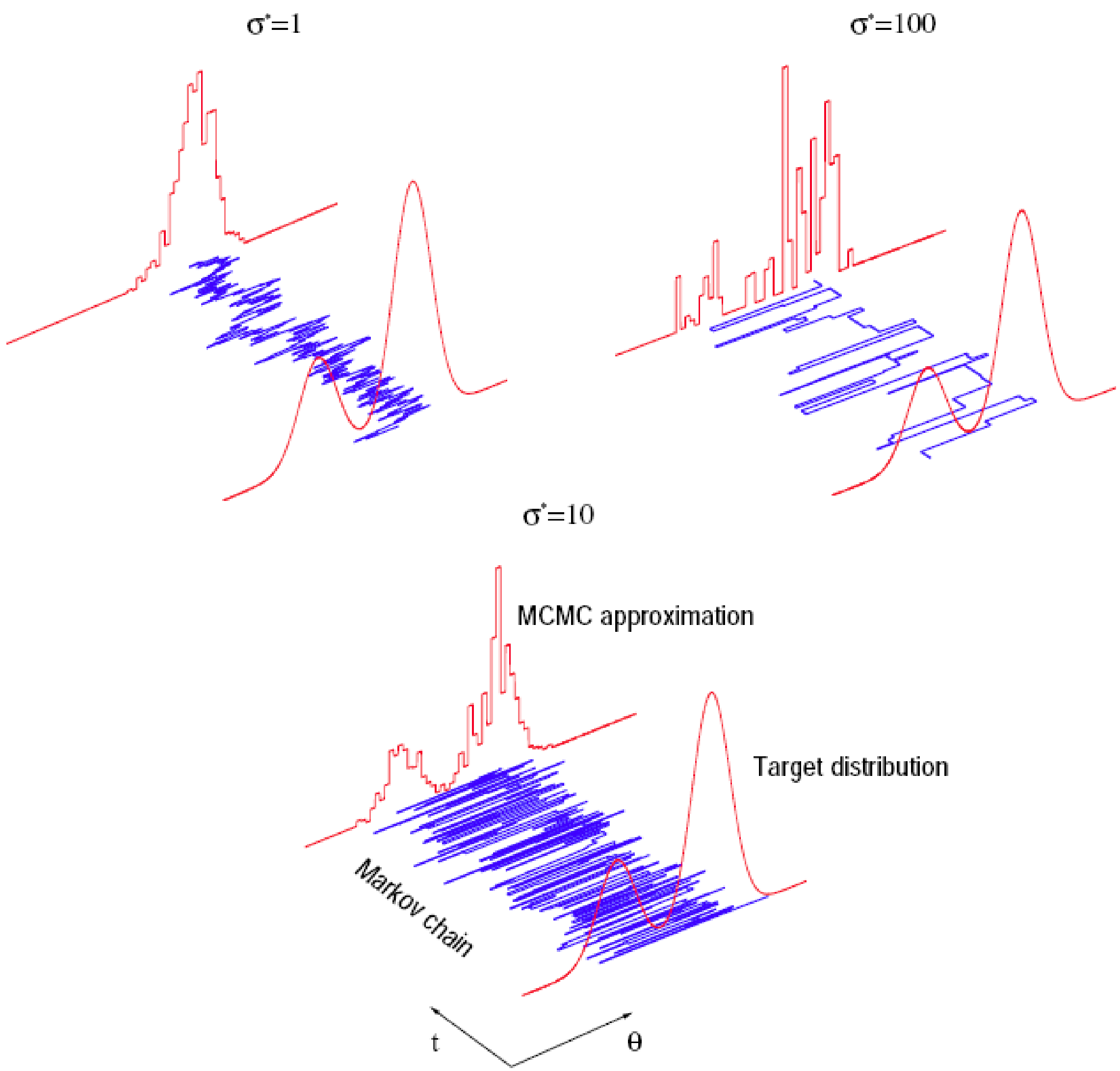
Importance of the proposal



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